

CLOUD ENGINEERING LAB · STUDENT PROJECT

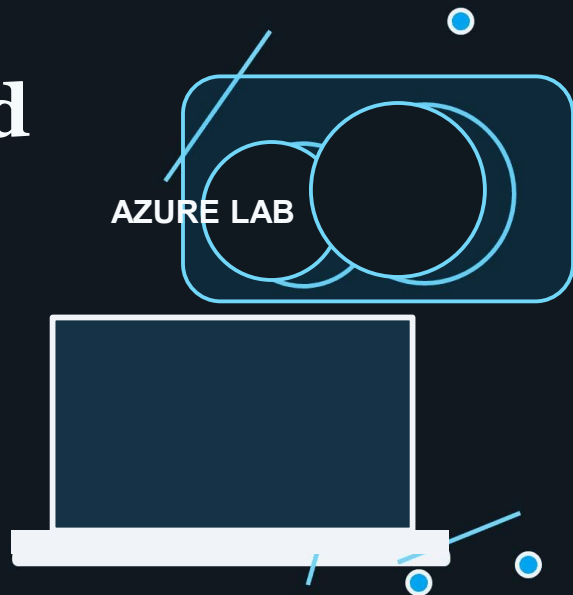
Azure Virtual Desktop Pooled Host Pool for a Lab

Cloud-Based Virtual Engineering Laboratory for Students

Team of 4 students

[KL University] · [Microsoft Azure]

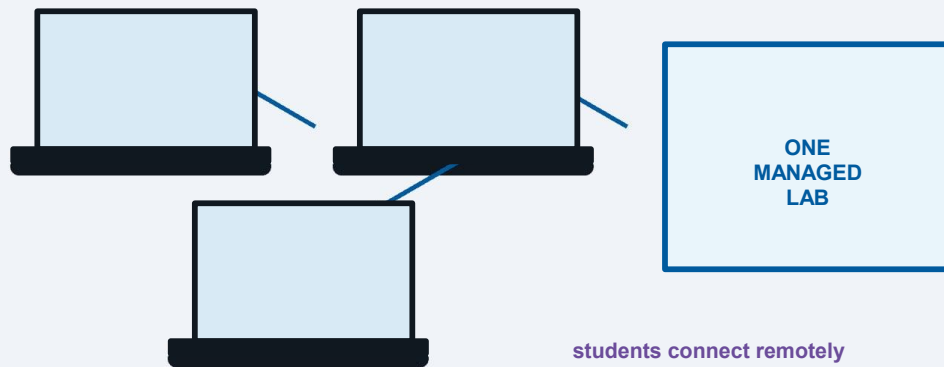
PLANNED IMPLEMENTATION & TESTING



One lab, many student devices

We need to give about 50 students access to engineering software—even when their personal laptops are not equally powerful.

- Different CPU, RAM, and graphics capability
- Local installation would be hard to maintain
- A shared cloud environment is the practical direction

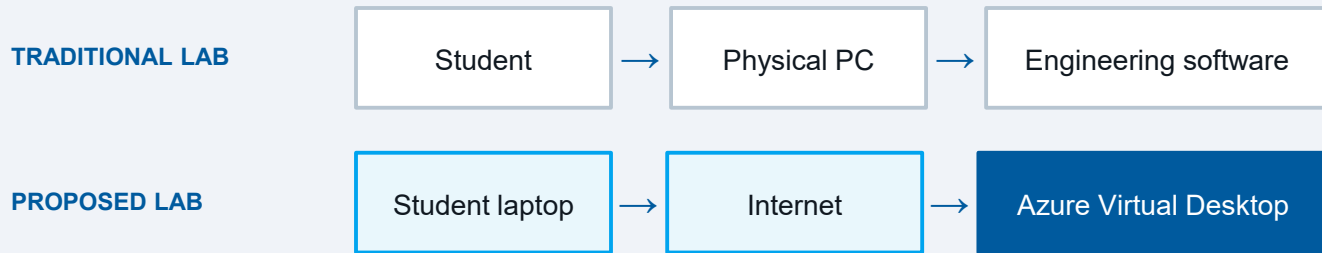


How a Student Will Use Our System

- **Step 1:** Student opens the Windows App on their personal laptop.
- **Step 2:** Student signs in securely with their college credentials.
- **Step 3:** Student clicks the "Virtual Lab" desktop icon.
- **Step 4:** AVD automatically connects them to an available Session Host VM.
- **Step 5:** The student opens the engineering software and completes their assignment.
- **Step 6:** Student saves work, logs out, and the host is freed for someone else.

Our solution moves the heavy work to Azure

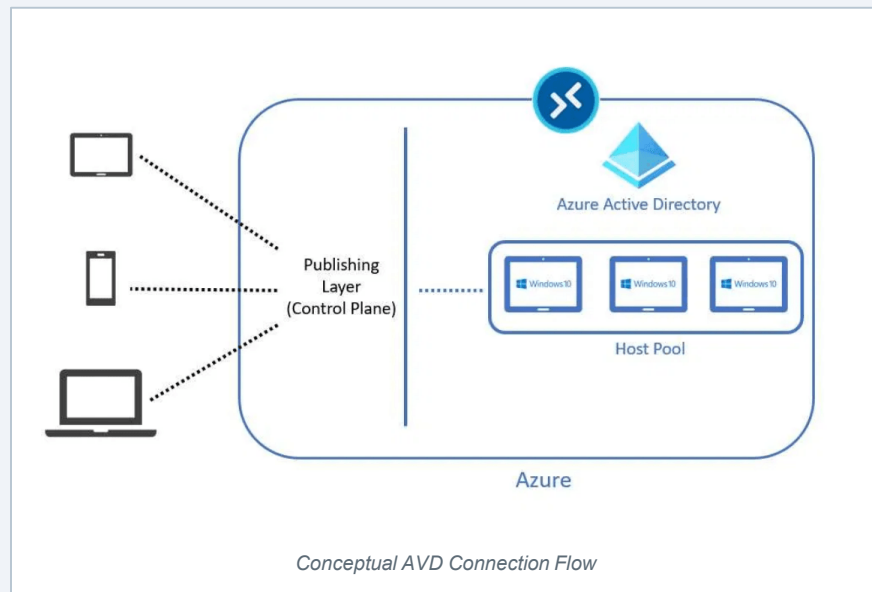
The laptop stays familiar. The computing environment moves.



“We do not want every student’s laptop to do the heavy work. We want Azure to provide the computing environment.”

THE LAPTOP BECOMES THE ACCESS DEVICE

What we plan to build



Core Components

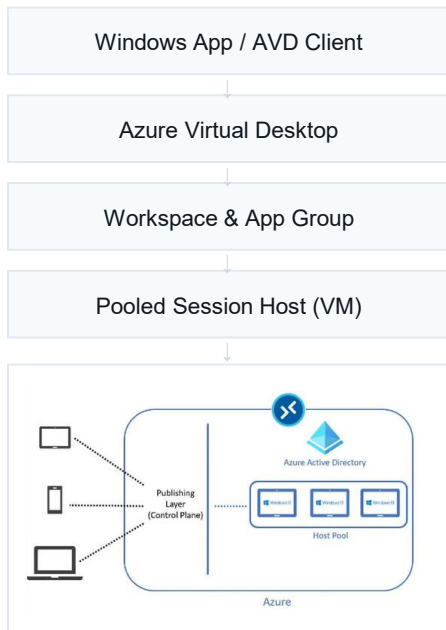
-  **Azure Virtual Desktop Workspace:** The student's entry point.
-  **Pooled Host Pool:** Shared session-host capacity.
-  **Session Host VMs:** Running the engineering software.
-  **Desktop Application Group:** Publishing the lab desktop.

User Profile Path

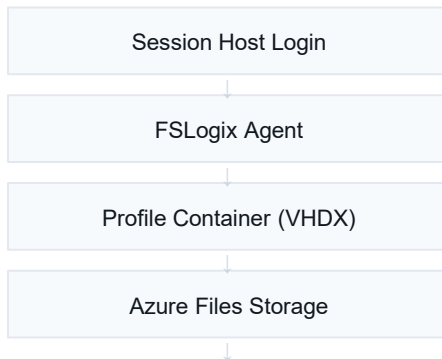
FSLogix + Azure Files: Profile containers are stored centrally so settings follow students across different hosts.

The Azure architecture in plain English

1. User Access Path

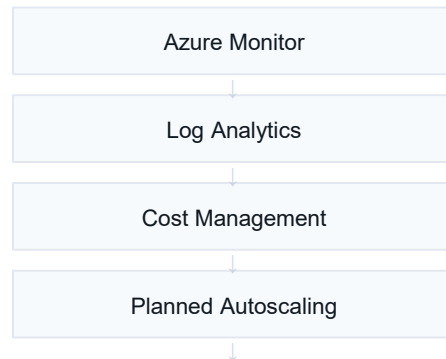


2. User Profile Path



Ensures student settings and files follow them across different VMs in the pool.

3. Operations & Scaling



Tracks performance, student-credit usage, and adjusts capacity based on demand.

We will build in stages, not all at once

STAGE 5: ESTIMATE

Capacity for ~50 students — design from results.

STAGE 4: MEASURE

Performance and cost — record evidence from Azure.

STAGE 3: LOAD TEST

Multiple test users — create realistic concurrent load.

STAGE 2: EXPAND

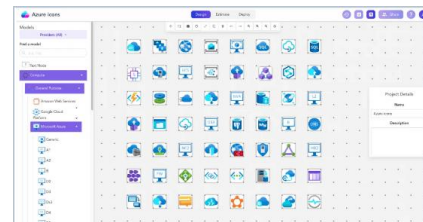
2–3 session hosts — observe pooled behavior.

STAGE 1: PROVE

1 session host — prove the basic connection.

CREDIT GUARDRAIL

Starting small protects our Azure for Students credit account and reduces avoidable risk.



Standard Azure Component Library

A staged build ensures we learn how the VM configuration behaves before scaling.

Implementation roadmap

STEP 01

Verify Azure for Students subscription

STEP 03

Set up Identity & User Access

STEP 05

Create first Session Host VM

STEP 07

Assign test users & connect from laptop

STEP 09

Configure FSLogix & Azure Files

STEP 02

Create Resource Group & Virtual Network

STEP 04

Create Pooled AVD Host Pool

STEP 06

Create Workspace & App Group

STEP 08

Add additional Session Hosts

STEP 10

Record configuration & observations

Delivering Engineering Software

- The core goal is remote delivery of heavy software (e.g., MATLAB, AutoCAD, SolidWorks).
- *Note: Specific software depends on our college's available licenses, which we will verify before deployment.*
- **Why centralize?** We maintain one single golden image. All students get the exact same software version.
- Students bypass the painful process of local installation and troubleshooting on underpowered laptops.



Bottleneck 1: sizing the session hosts

Key performance variables:

CPU Cores

RAM Capacity

Disk Speed

Network Latency

GPU (Graphics)



Bottleneck 1: Session Host Sizing

The Challenge: Session host sizing for graphics workloads is hard to estimate.

In our own words: We don't know beforehand how powerful one Azure VM needs to be for several students using 3D engineering software simultaneously.

- **Too small:** Software lags, poor student experience.
- **Right size:** Acceptable performance, balanced cost.
- **Too large:** Wastes limited student Azure credits.



How We Will Solve Session Host Sizing



The Experiment

Test 1 → 1 user

Test 2 → 3 users

Test 3 → 5 users

(More if budget permits)



What We Measure

CPU and Memory usage

GPU usage (if applicable SKU)

App startup time

General responsiveness



The Outcome

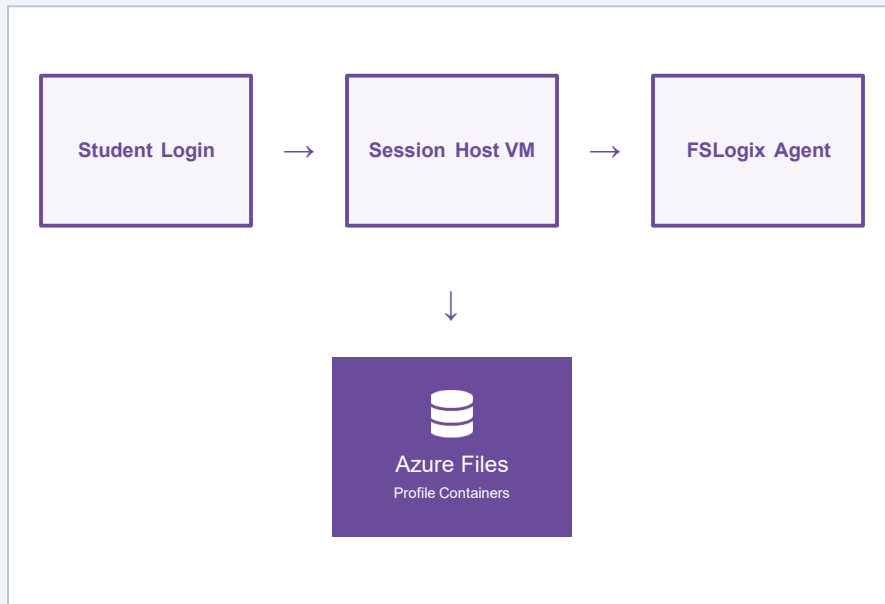
Our final VM size recommendation will be based on **[Actual Result]** measurements, not guesswork.

Bottleneck 2: FSLogix and Azure Files

The Pooled-Profile Challenge

In a pooled host pool, students connect to different VMs each time. We need their settings and files to follow them seamlessly.

- ☁ **FSLogix:** Provides a profile-container approach that mounts the user profile at login.
- ☁ **Azure Files:** Acts as the central storage for these profile containers.
- ☁ **Login Experience:** Storage performance directly impacts how fast a student can start working.

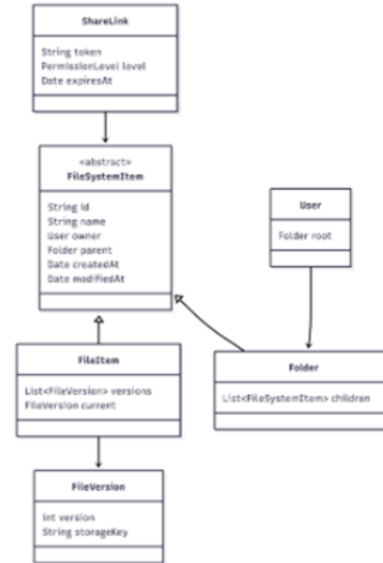


FSLogix Storage

The Challenge: FSLogix profile container storage can become a bottleneck.

Because we use a Pooled Host Pool, a student might log into VM01 on Monday and VM03 on Tuesday. Their settings and files must follow them.

FSLogix solves this by storing profiles in Azure Files. However, if 20 students log in at the exact same minute, the storage network can become overwhelmed, causing massive login delays.



How We Will Test the FSLogix Bottleneck

Storage Experiment

We will test increasing concurrent user logins to simulate a lab session starting:

- 1 concurrent login
- 5 concurrent logins
- 10 concurrent logins
- [To be tested] maximum allowed by our environment.

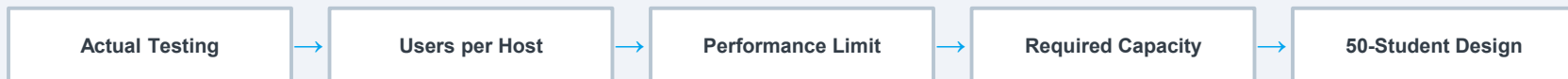
What We Measure

We will use Azure metrics to monitor:

- Total login time (in seconds)
- Profile loading time
- Storage latency and disk IO delays

If profile storage becomes slow, we will investigate Azure Files tier configurations to solve it.

From measurements to a 50-student design



Measured Users per Host

[Actual Result]

Required Session Hosts

[To be calculated]

Target Concurrency

~50 Students

Our final design will distinguish between **enrolled students** and **simultaneous users**. By using Azure Monitor and Cost Management, we will scale the environment based on evidence rather than assumptions.

We will not simply guess that "X VMs support 50 students." We will use our experimental data to calculate it.



1. Actual Testing

Run load tests on a single Session Host VM.



2. Measure Limit

Determine **[Actual Result]** maximum users per host before lag occurs.



3. Calculate Needs

Divide 50 concurrent students by our safe user-per-host limit.



4. Final Design

Propose a pool of **[Calculated]** Session Hosts to support the lab safely.

Our experiments will produce the evidence

Track 1: Session Host Sizing

- Test 1: 1 concurrent user
- Test 2: 3 concurrent users
- Test 3: 5 concurrent users

VM CONFIG: [To be measured]
ACCEPTABLE LIMIT: [Actual result]

Track 2: Profile Storage (FSLogix)

- Test 1: 1 concurrent login
- Test 2: 5 concurrent logins
- Test 3: 10+ concurrent logins

STORAGE CONFIG: [To be measured]
PEAK OBSERVED DELAY: [Actual result]

RESOURCE USAGE

CPU · RAM · GPU

USER EXPERIENCE

Responsiveness

TIME METRICS

Login · App Startup

STORAGE HEALTH

Latency · Disk I/O



Our Final Plan Summary

- ✓ **BUILD:** A pooled AVD environment in Azure for personal device access.
- ✓ **TEST:** Session-host sizing for heavy engineering software.
- ✓ **INVESTIGATE:** FSLogix and Azure Files storage bottlenecks.
- ✓ **OPTIMIZE:** Use Azure Monitor and autoscaling for cost control.
- ✓ **SCALE:** Use measured data to design the final 50-student lab.

Thank You!